

K-FIELD

Compton-Classical-Radius Mass-Bridge Triangle

A geometric and counter-rotating synthesis of electron, proton,
electromagnetic, and Earth-CMB phase scales

Principal Investigator: P. Dwayne Esterline

BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)

Krytronx, LLC

Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

Archival publication epoch: 1785175483

Generated UTC: 2026-07-27T18:04:43.208974Z

Coordinate frame: ICRS/J2000 Cartesian for the Earth-CMB vector

Cell scale: 16.0686029005 nm per native unit

Foundational statement

The three Mass-Bridge ratios form one closed physical triangle: electron reduced Compton scale, classical electron radius, and proton reduced Compton scale. The Earth-CMB traversal supplies a common de Broglie phase factor, while counterrotation separates even phase energy from odd angular momentum and lateral response.

Proprietary Notice and Document Contract

Owner: Krytronx, LLC. Principal Investigator: P. Dwayne Esterline. This document contains proprietary confidential information and trade secret material. Distribution, reproduction, disclosure, or derivative use is prohibited except by explicit authorization of Krytronx, LLC.

This paper is a standalone foundational education and archival reference. Numerical tables and plotted quantities are generated directly from the stated Cell3 geometry, the stated Earth-CMB vector and K-Field values, and the CODATA 2022 constant table distributed with SciPy 1.17.0. Equations are separated into exact known-physics identities, direct corpus evaluations, structural correspondences, and exploratory composition laws.

Abstract

The Mass-Bridge geometry contains two primary area ratios and one derived cross-ratio. Direct evaluation gives $1000 A_{bc}/A_{ab} = 1836.152689330240$, $100 A_{bc}/A_{ac} = 137.036000351759$, and $10 A_{ac}/A_{ab} = 13.399053421123$. These map respectively to the proton-electron mass ratio η , the inverse fine-structure constant α^{-1} , and $\eta \alpha$. The corresponding known-physics scales are the electron reduced Compton wavelength, classical electron radius, and proton reduced Compton wavelength. Their ratios obey exact identities, so the three geometric bridge relations are not independent numerical targets but one closed projective triangle.

The Earth-CMB speed produces a common de Broglie-to-Compton ratio $1/(\gamma_E \beta_E)$ shared by the electron and proton. The K-Field directional factor maps this common traversal scale into the neighborhood of $\eta/\sqrt{11}$. Counter-rotating balance then separates the equal-phase-gradient, equal-energy, and zero-angular-momentum states because the K-response channel is weighted by $m \omega$ while angular momentum is weighted by $m \rho^2 \omega$. The resulting hierarchy explains why the electron-inner/proton-outer ordering is physically coherent in reciprocal phase space rather than as a literal direct-space shell model.

Central result

$A_{bc}:A_{ac}:A_{ab}$ is projectively equivalent to $\lambda_{\bar{e}}:(100 r_e):(1000 \lambda_{\bar{p}})$. The shared face cancels in the cross-ratio, producing $10 A_{ac}/A_{ab} = r_e/\lambda_{\bar{p}} = \eta \alpha$ to approximately 89 parts per trillion relative precision.

Contents

Proprietary Notice and Document Contract

2

Abstract

2

1. Scope, notation, and evidentiary layers

5

Four evidentiary layers

5

2. Known-physics foundation

5

2.1 Rest energy and Compton frequency

5

2.2 Reduced and ordinary Compton wavelengths

5

2.3 de Broglie wavelength

5

2.4 Classical electron radius and alpha

6

3. Cell3 base geometry

6

3.1 The 27 centered half-step nodes

7

4. Cell3 face planes and Mass-Bridge observables

8

5. The Compton-classical-radius Mass-Bridge triangle

11

5.1 Exact physical triangle

11

6. Common-normalization structure

13

7. Direct-space and reciprocal-space interpretation

14

8. Earth-CMB de Broglie bridge

14

8.1 K-Field directional scaling and the sqrt(11) neighborhood

15

9. Counter-rotating parity and balance

15

9.1 Quadratic phase energy and inertial curvature

15

9.2 Differential-pair equations

16

10. Earth-CMB vector and directional response

17

10.1 Secondary 10/9 observation

18

11. Exploratory two-body correction stack

18

12. Hidden geometric connections and derived consequences

19

12.1 One bridge, not two coincidences

19

12.2 The electromagnetic node is the bridge pivot

20

12.3 Direct and reciprocal duality

20

12.4 CMB translation is common-mode before geometry 20

12.5 Counterrotation separates parity sectors 20

12.6 Experimental null architecture 20

13. Consolidated conclusions 20

Appendix A. Complete numerical constant table 22

Appendix B. Complete 27-node coordinate table 23

Appendix C. Compact derivation set 24

 C.1 Known-physics triangle 24

 C.2 Mass-Bridge triangle 24

 C.3 Earth-CMB common factor 24

 C.4 Counter-rotating balance 24

 C.5 Quadratic parity cancellation 24

Appendix D. Reproducibility record 24

References 24

1. Scope, notation, and evidentiary layers

This paper develops a unified geometric interpretation of four bodies of information: standard relativistic and quantum relationships for the electron and proton; the Cell3 base geometry and its three face areas; the measured Earth-CMB velocity used by the K-Field corpus; and the parity-separating equations of a concentric counter-rotating pair. The purpose is to identify relations that are algebraically exact, relations that are direct numerical consequences of the corpus, and relations that remain composition hypotheses.

The notation uses $\eta = m_p/m_e$ for the proton-electron mass ratio, α for the fine-structure constant, c for the speed of light, h for the Planck constant, and $\hbar = h/(2\pi)$. A bar over a Compton wavelength is written as λ_{bar} in ASCII equations. The rotor radii ρ_e and ρ_p are generalized phase radii. They need not be ordinary spatial shell radii.

Four evidentiary layers

Exact known-physics identities. These follow algebraically from definitions such as $\lambda_{\text{bar}} = \hbar/(m c)$ and $r_e = \alpha \hbar/(m_e c)$.

Direct corpus evaluations. These are computed from the supplied Cell3 lengths, angles, face areas, Earth-CMB speed, transformed K-Field vector, and directional factor.

Structural correspondences. These map a set of corpus ratios to a set of physical ratios while preserving all pairwise relations.

Exploratory composition laws. These combine independent-looking factors to test whether a closed derivation is available. They are retained only when every factor has a stated physical role and the remaining residual is explicitly reported.

2. Known-physics foundation

Rest energy, Compton scale, de Broglie scale, and electromagnetic radius are different expressions of mass, phase, and coupling. Their relationships provide the reference frame for evaluating the Mass-Bridge.

2.1 Rest energy and Compton frequency

$$E_0 = m c^2 \text{ and } \nu_C = E_0 / h = m c^2 / h$$

For two particles evaluated with the same constants, rest energy and Compton frequency scale linearly with mass. Therefore $E_{0,p}/E_{0,e} = \nu_{C,p}/\nu_{C,e} = \eta$. These are equivalent mass statements rather than statistically independent measurements.

2.2 Reduced and ordinary Compton wavelengths

$$\lambda_{\text{bar}} = \hbar/(m c) ; \lambda_C = h/(m c) = 2\pi \lambda_{\text{bar}}$$

The reduced Compton wavelength is the natural radius of angular phase because $\hbar$ is the quantum of angular action. The ordinary Compton wavelength is the length accumulated over a full 2π phase cycle. The factor 2π therefore separates angular circulation from linear wavelength closure; it is not an arbitrary registry multiplier.

2.3 de Broglie wavelength

$$\lambda_{\text{dB}} = h/p = h/(\gamma m v)$$

At a common velocity v , the de Broglie wavelength is inversely proportional to mass. The electron-to-proton wavelength ratio is η . Dividing by the ordinary Compton wavelength removes mass entirely and leaves $1/(\gamma \beta)$, where $\beta = v/c$.

2.4 Classical electron radius and alpha

$$r_e = e^2 / (4 \pi \epsilon_0 m_e c^2) = \alpha \hbar / (m_e c) = \alpha \lambda_{\text{bar}_e}$$

The classical electron radius is the electromagnetic length obtained by comparing the electron charge coupling with its rest-energy scale. It lies between the proton and electron reduced Compton wavelengths. This intermediate position is the key to the three-node bridge.

Quantity	Definition	Value
eta	m_p/m_e	1836.152673426000
alpha^-1	$1/\alpha$	137.035999177590
lambda_bar_e	$\hbar/(m_e c)$	3.861592674352e-13 m
r_e	$\alpha \lambda_{\text{bar}_e}$	2.817940320447e-15 m
lambda_bar_p	$\hbar/(m_p c)$	2.103089100514e-16 m

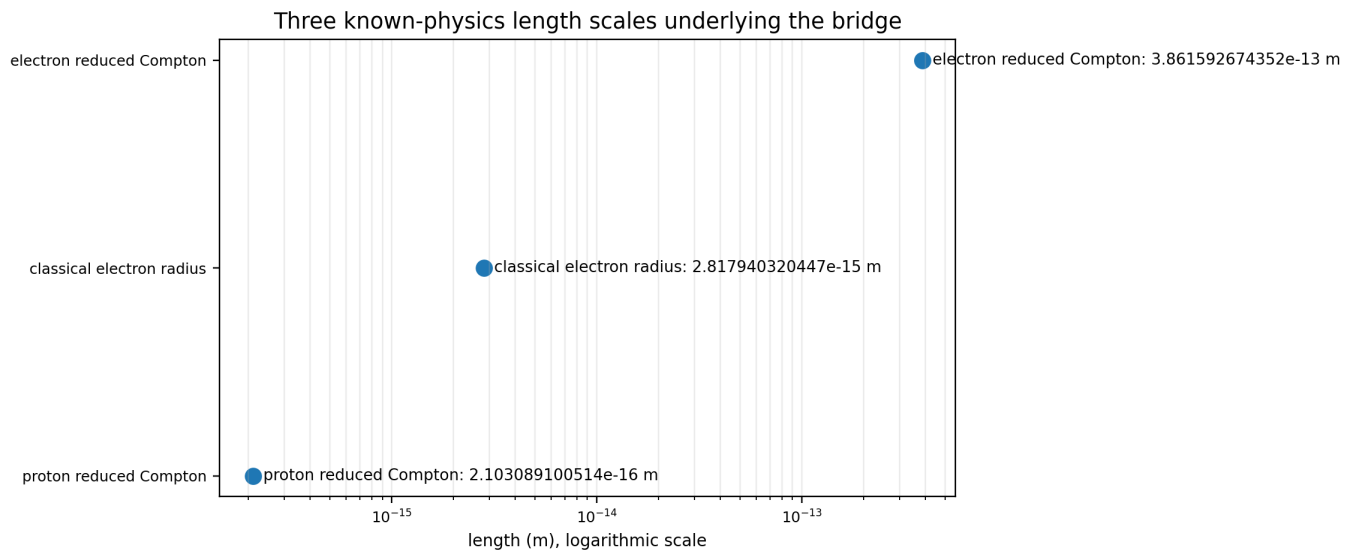


Figure 1. The electron reduced Compton wavelength, classical electron radius, and proton reduced Compton wavelength on a dimensionally labeled logarithmic length axis.

3. Cell3 base geometry

The Cell3 base cell is defined by three nonorthogonal edge vectors a , b , and c . Their measured magnitudes and pairwise angles uniquely determine the cell up to rigid rotation and reflection. A canonical Cartesian reconstruction places a on the x axis, b in the x - y plane, and c above that plane.

$$a = (|a|, 0, 0) \quad b = (|b| \cos \theta_{ab}, |b| \sin \theta_{ab}, 0) \quad c_x = |c| \cos \theta_{ac} \quad c_y = |c| (\cos \theta_{bc} - \cos \theta_{ab} \cos \theta_{ac}) / \sin \theta_{ab} \quad c_z = \sqrt{|c|^2 - c_x^2 - c_y^2}$$

The reconstruction is not an additional model assumption. It is a coordinate realization of the supplied Gram matrix. All lengths, angles, areas, and volume are preserved.

Geometric quantity	Native value	Physical value
a	0.0207970587	0.3341796777 nm
b	0.0364942052	0.5864108915 nm
c	0.0493085660	0.7923197666 nm
theta_ab	83.933483 deg	same angle
theta_ac	80.448130 deg	same angle
theta_bc	50.363232 deg	same angle

Geometric quantity	Native value	Physical value
cell volume	2.842066945211e-05 native^3	1.179148922239e-01 nm^3

Canonical Cell3 base cell reconstructed from measured lengths and angles

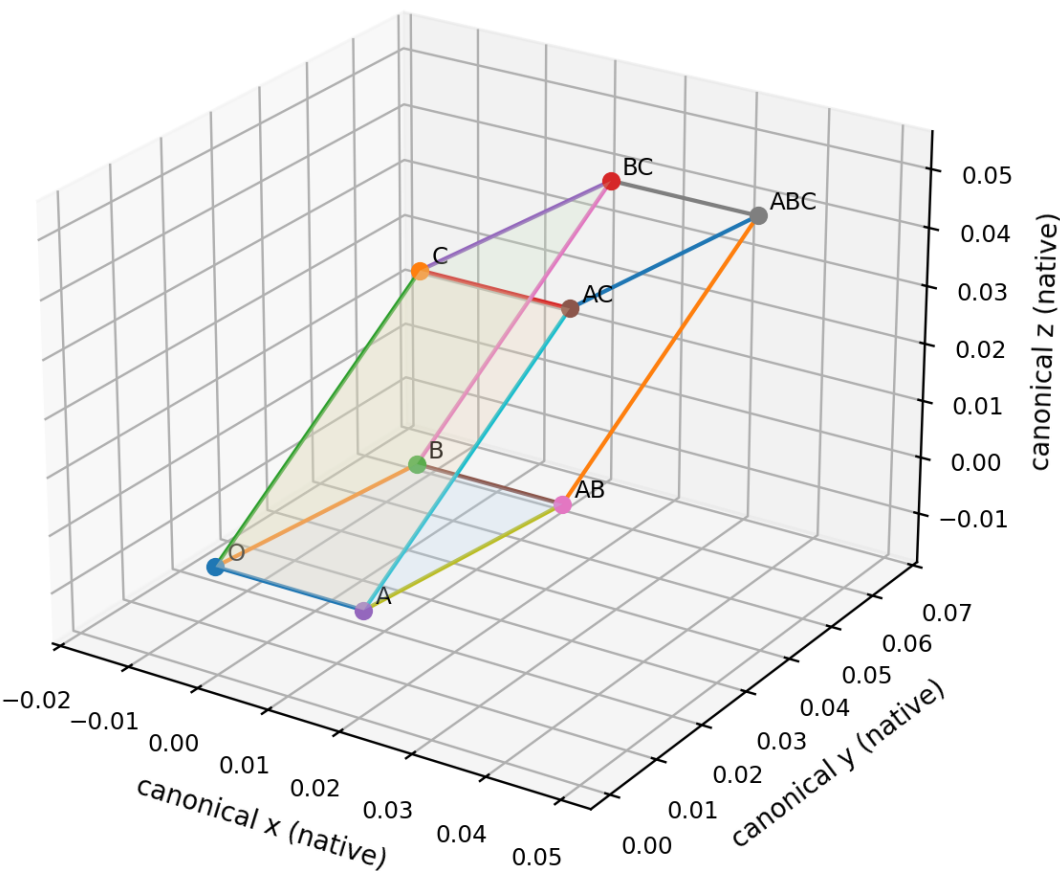


Figure 2. Dimensionally faithful canonical Cell3 base cell. The eight vertices are O, A, B, C, AB, AC, BC, and ABC. Axis units are native Cell3 units and all three axes use the same scale.

3.1 The 27 centered half-step nodes

The centered Cell3 node set consists of the coordinates $(i a + j b + k c)/2$ for i, j, k in $\{-1,0,+1\}$. It contains eight corners, twelve edge centers, six face centers, and one body center. This is the complete 3 by 3 by 3 address set.

Node count identity
8 corners + 12 edge centers + 6 face centers + 1 body center = 27 nodes.

Physical Cell3: all 27 centered half-step nodes

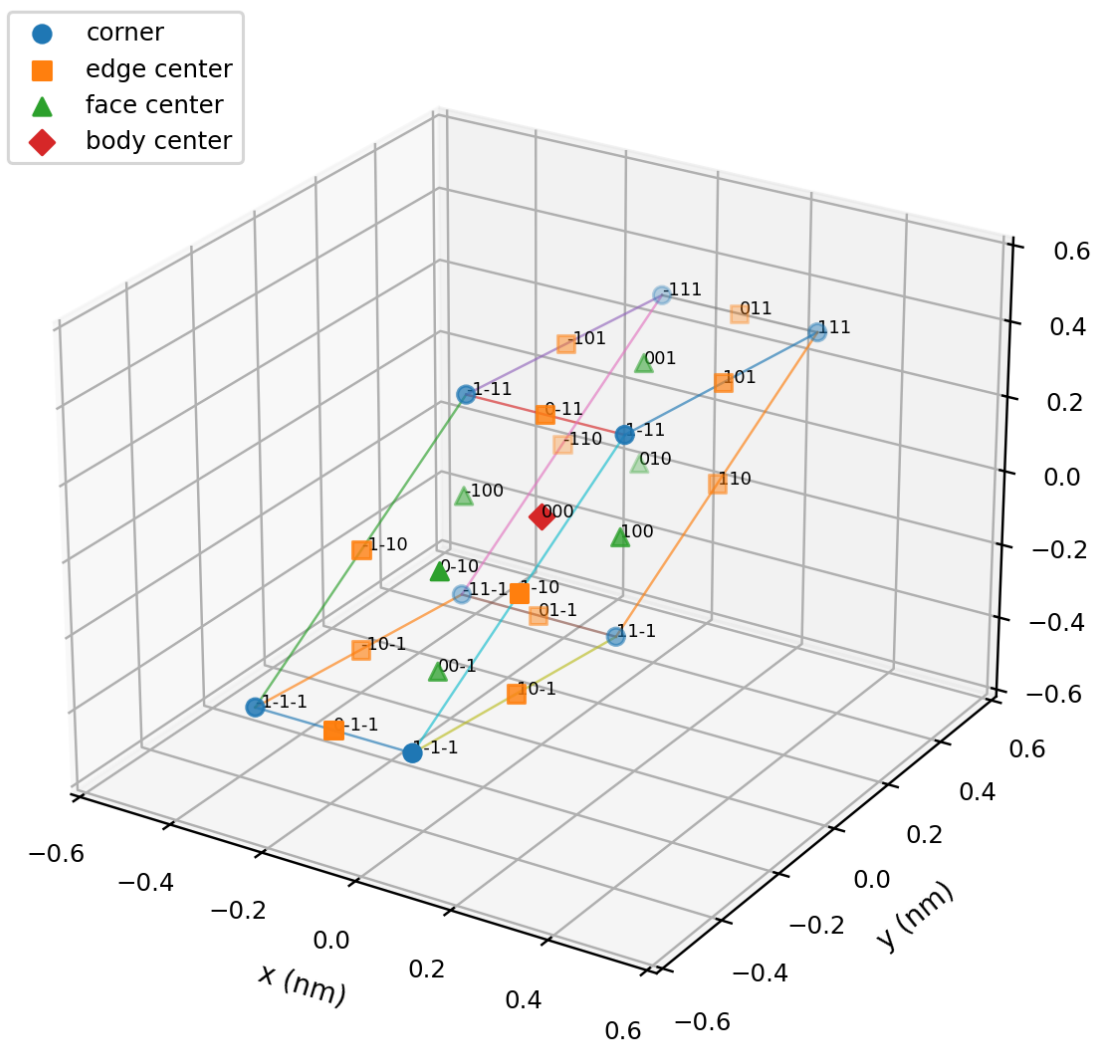


Figure 3. All 27 centered half-step nodes at the corpus physical scale. Each label is the integer address ijk, and each axis is in nanometres with equal three-dimensional scaling.

4. Cell3 face planes and Mass-Bridge observables

Each pair of edge vectors defines a physical parallelogram face. Its area is the magnitude of the corresponding cross product. The three face areas are the primary geometric observables used by the Mass-Bridge.

$$A_{ab} = |a \times b| = |a||b| \sin \theta_{ab} \quad A_{ac} = |a \times c| = |a||c| \sin \theta_{ac} \quad A_{bc} = |b \times c| = |b||c| \sin \theta_{bc}$$

Face	Nodes	Native area	Physical area	Unit normal in canonical frame
ab	O, A, B, AB	7.547217880965119e-04	0.194869165063 nm ²	+0.000000000, +0.000000000, +1.000000000
ac	O, A, C, AC	1.011255755679060e-03	0.261106235281 nm ²	+0.000000000, -0.774440982, +0.632646161
bc	O, B, C, BC	1.385784440909538e-03	0.357809541498 nm ²	+0.986135754, -0.104804678, -0.128655566

ab face plane: area = 0.194869165 nm²

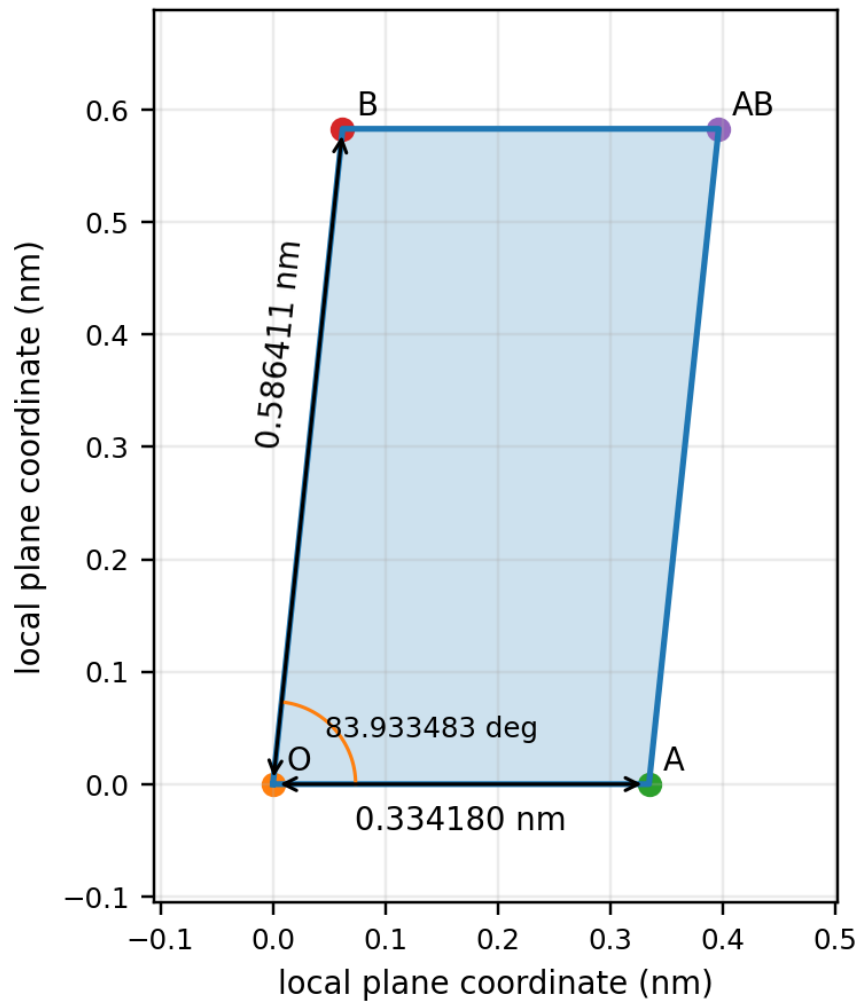


Figure 4. The ab face plane with actual physical edge lengths, included angle, node names, and physical area. The drawing uses equal local x and y scales.

ac face plane: area = 0.261106235 nm^2

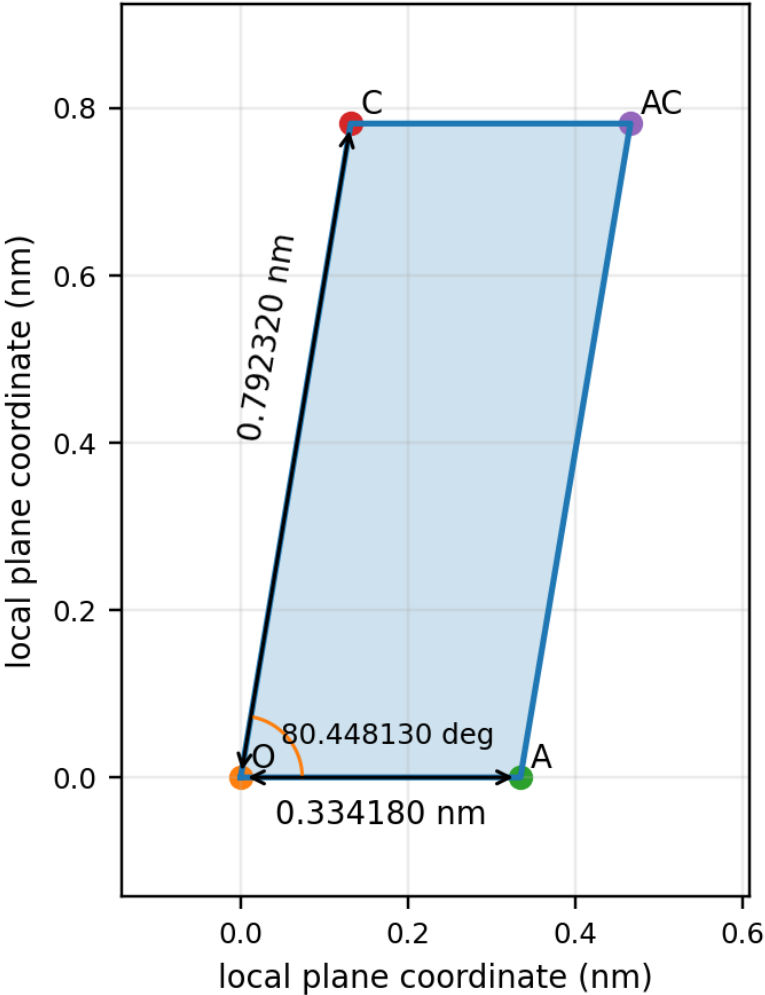


Figure 5. The ac face plane with actual physical edge lengths, included angle, node names, and physical area. The drawing uses equal local x and y scales.

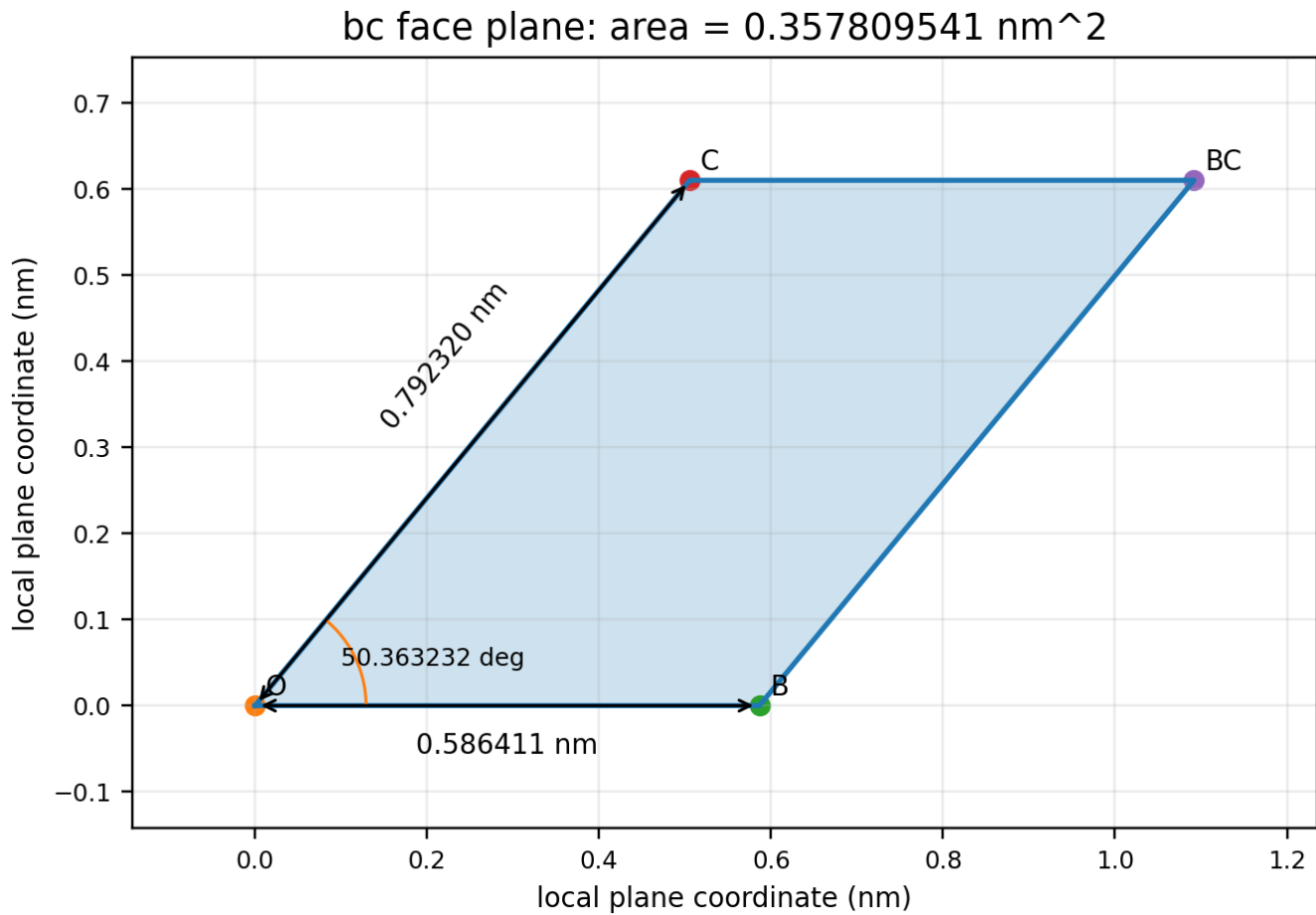


Figure 6. The bc face plane with actual physical edge lengths, included angle, node names, and physical area. The drawing uses equal local x and y scales.

5. The Compton-classical-radius Mass-Bridge triangle

The three face-area expressions generate two primary bridge ratios and a cross-ratio in which the shared face A_{bc} cancels.

$$\eta_{\text{geom}} = 1000 A_{bc}/A_{ab} \quad \alpha_{\text{inv_geom}} = 100 A_{bc}/A_{ac} \quad \chi_{\text{geom}} = 10 A_{ac}/A_{ab} = \eta_{\text{geom}}/\alpha_{\text{inv_geom}}$$

Direct evaluation gives:

Bridge quantity	Cell3 value	Known-physics value	Signed relative residual
1000 A_{bc}/A_{ab}	1836.152689330240	$\eta = 1836.152673426000$	+8.662 ppb
100 A_{bc}/A_{ac}	137.036000351759	$\alpha^{-1} = 137.035999177590$	+8.568 ppb
10 A_{ac}/A_{ab}	13.399053421123	$\eta \alpha = 13.399053419872$	+93.4 ppt

5.1 Exact physical triangle

$$\lambda_{\text{bar}_e}/\lambda_{\text{bar}_p} = m_p/m_e = \eta \quad \lambda_{\text{bar}_e}/r_e = \alpha^{-1} \quad r_e/\lambda_{\text{bar}_p} = \eta \alpha$$

The first identity follows because reduced Compton wavelength is inversely proportional to mass. The second follows from $r_e = \alpha \lambda_{\text{bar}_e}$. The third is the product of the first and the inverse of the second. The Mass-Bridge therefore closes on one three-node physical scale system rather than two unrelated constants.

Projective correspondence
 $A_{bc} : A_{ac} : A_{ab} \leftrightarrow \lambda_{\bar{e}} : (100 r_e) : (1000 \lambda_{\bar{p}})$.
The factors 100 and 1000 are registry placements. The unscaled physical content is $\lambda_{\bar{e}} : r_e : \lambda_{\bar{p}}$.

Compton-classical-radius Mass-Bridge triangle

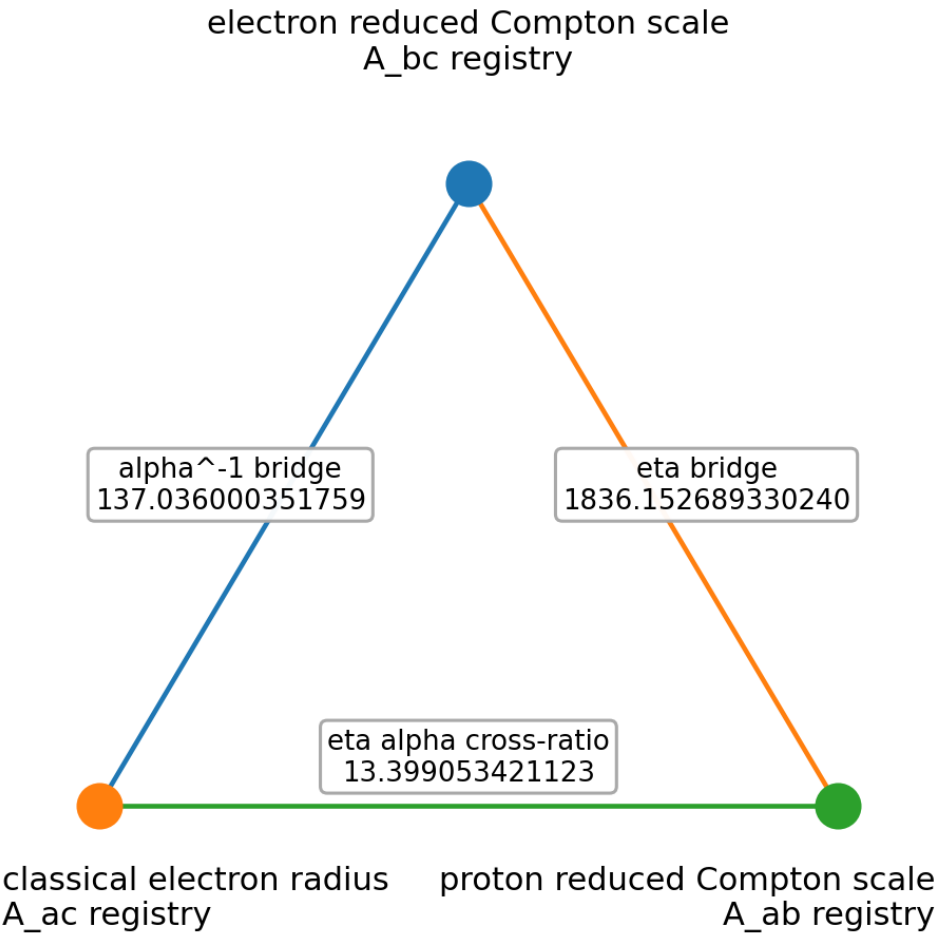


Figure 7. The closed Mass-Bridge triangle. Every edge has both a Cell3 area-ratio expression and an exact known-physics length-ratio expression.

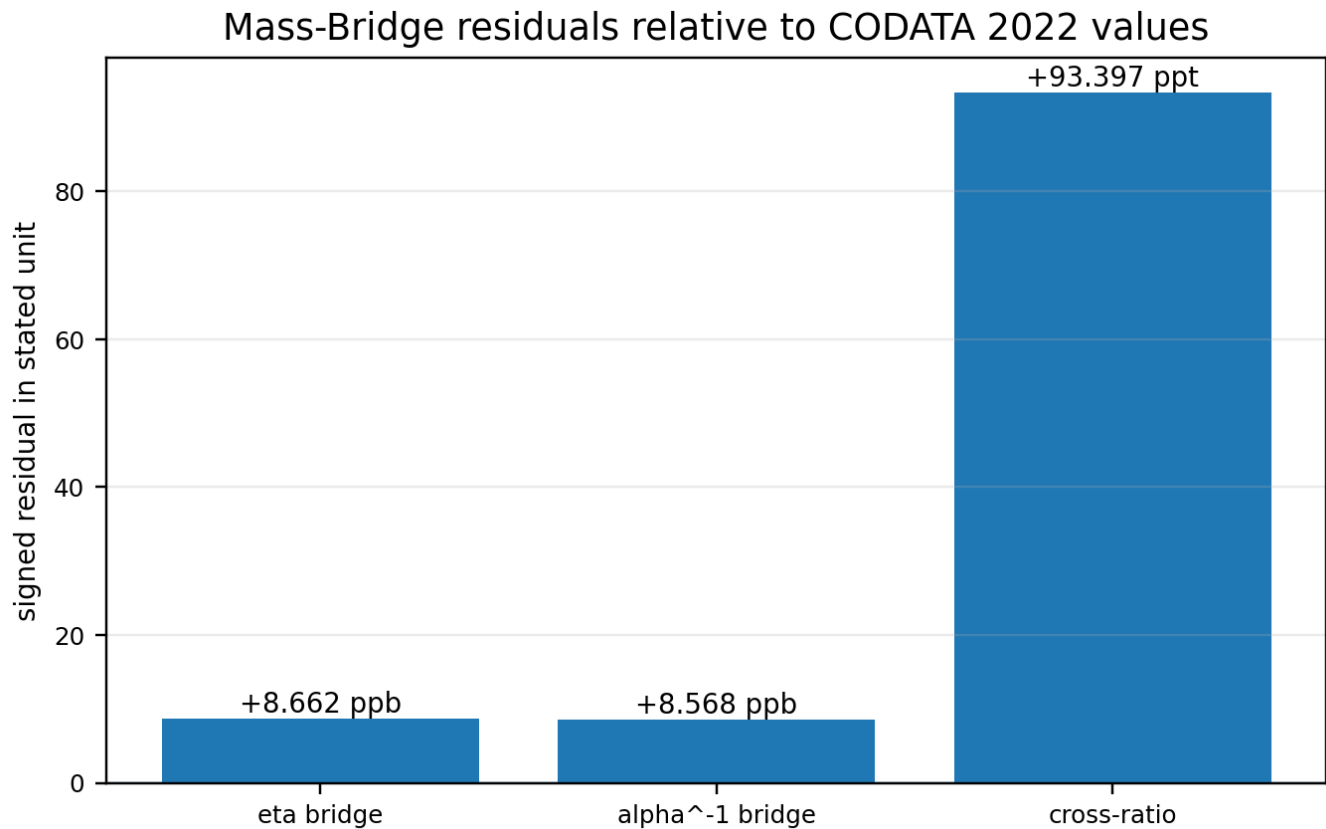


Figure 8. Signed numerical residuals. The cross-ratio is displayed in parts per trillion because the shared A_{bc} normalization cancels.

6. Common-normalization structure

The eta and α^{-1} bridge relations both contain A_{bc} in the numerator. A common multiplicative displacement of A_{bc} therefore moves both ratios together without changing the cross-ratio. The scale factors required to match the two known constants independently are

```
s_eta = eta/eta_geom = 0.9999999913382802 s_alpha = alpha_inv/alpha_inv_geom =
0.9999999914316772
```

Their geometric mean is

```
s_common = sqrt(s_eta s_alpha) = 0.9999999913849787
```

Applying this one factor to A_{bc} leaves equal-and-opposite relative residuals of approximately $4.670e-11$ and $-4.670e-11$. The near symmetry demonstrates that almost all of the remaining primary-ratio mismatch behaves as a common A_{bc} normalization term. The cross-ratio is unaffected because A_{bc} is absent.

This separation is important. The intrinsic Cell3 shape is tested by $10 A_{ac}/A_{ab}$. The shared normalization of the two primary bridges is tested by A_{bc} . Shape and common scale are therefore independently observable.

7. Direct-space and reciprocal-space interpretation

In ordinary hydrogen center-of-mass coordinates, the proton occupies the inner spatial region and the electron occupies the outer spatial region. The requested electron-inner/proton-outer ordering emerges when the radius is a reciprocal phase coordinate rather than an ordinary distance.

Define the reduced-Compton wavenumber

$$k_C = 1/\lambda_{\text{bar}} = m c / \hbar$$

Then

$$k_{C,p}/k_{C,e} = m_p/m_e = \eta$$

The electron is inner and the proton is outer along a reciprocal-radius axis. Rest energy, Compton frequency, reciprocal radius, and inverse reduced-Compton length all produce the same ordering:

$$E_{0,p}/E_{0,e} = \nu_{C,p}/\nu_{C,e} = k_{C,p}/k_{C,e} = \lambda_{\text{bar}_e}/\lambda_{\text{bar}_p} = \eta$$

These equalities are multiple representations of one mass scaling. Their value is structural: they show that the phase rotor picture is compatible with standard physics when the rotor coordinate is reciprocal.

Interpretive rule

The electron-inner/proton-outer pair is a reciprocal phase-space model. It should not be drawn as two literal mechanical shells with direct-space radii proportional to the Compton wavenumbers.

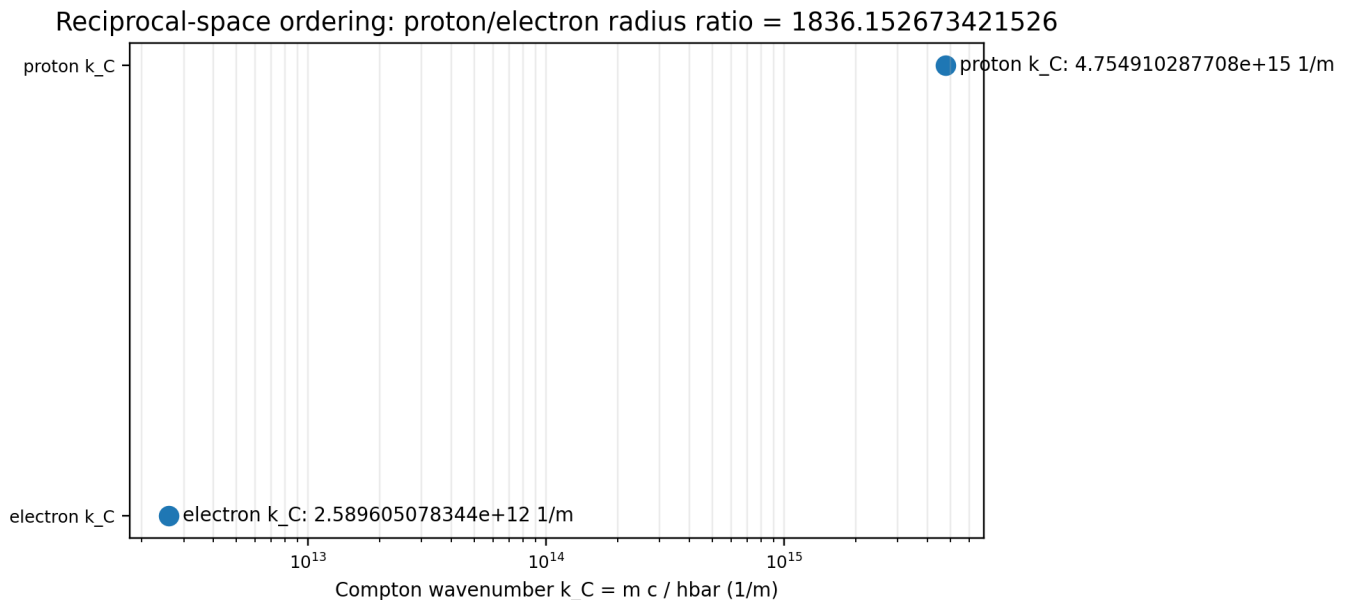


Figure 9. Dimensioned reciprocal-space ordering. The axis is the Compton wavenumber k_C in inverse metres; the proton lies outward by the factor η .

8. Earth-CMB de Broglie bridge

The Earth-CMB speed used by the corpus is $V_E = 369827.392$ m/s. The corresponding relativistic factors are $\beta_E = 0.001233611394$ and $\gamma_E = 1.000000760899$.

$$\lambda_{dB,i} = h / (\gamma_E m_i V_E)$$

Particle	Earth-CMB de Broglie wavelength	Ordinary Compton wavelength	λ_{dB}/λ_C
electron	1.966833640774e-09 m	2.426310235380e-12 m	810.627434239027
proton	1.071171079205e-12 m	1.321409853604e-15 m	810.627434239027

The electron-to-proton de Broglie wavelength ratio is

$$\lambda_{dB,e}/\lambda_{dB,p} = 1836.152673421527 = \eta$$

For either particle, mass cancels from the de Broglie-to-Compton ratio:

$$\lambda_{dB}/\lambda_C = 1/(\gamma_E \beta_E) = 810.627434239027$$

This is a universal translational phase factor created by the common Earth-CMB velocity. It acts equally on electron and proton phase populations before any K-Field directional weighting is applied.

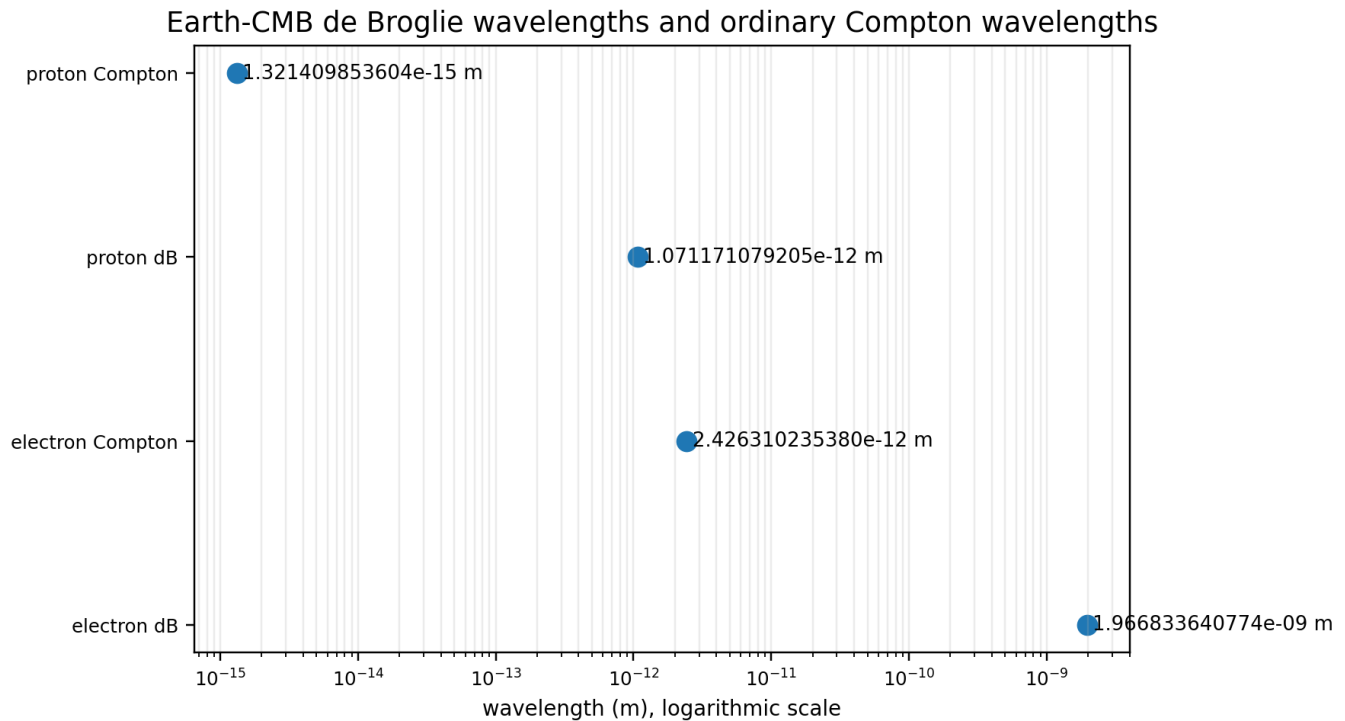


Figure 10. Earth-CMB de Broglie wavelengths and ordinary Compton wavelengths for the electron and proton, evaluated from the stated velocity and CODATA constants.

8.1 K-Field directional scaling and the sqrt(11) neighborhood

$$S_K = 1/(\gamma_E \beta_E f_K) = 553.762035958415$$

$$\text{sqrt}(11) S_K = 1836.620896417359$$

This value lies 255.002 parts per million above η . Using the separately reported $S_{CMB} = 553.784756$ gives $\text{sqrt}(11) S_{CMB} = 1836.696250270516$, which lies 296.041 parts per million above η .

The CMB metric and the discrete sqrt(11) mode therefore identify the correct mass-ratio neighborhood, but they do not by themselves reproduce the high-precision Mass-Bridge result. The remaining normalization is a distinct layer.

9. Counter-rotating parity and balance

A concentric counter-rotating pair is a physical parity and weighting separator. It independently resolves an even phase-energy channel, an odd angular-momentum channel, and an odd K-Field lateral-response channel.

9.1 Quadratic phase energy and inertial curvature

$$q_{\dot{}} = 2 B^{-1} v \Phi = q_{\dot{}}^{\dot{}} T W \quad q_{\dot{}} = v^{\dot{}} T G_{\text{phase}} v \beta(u) \Phi = v^2/2$$

After the directional normalization, $E = m v^2/2$, $p = dE/dv = m v$, and inertial mass is the curvature of the normalized phase-energy basin. Energy is even under velocity reversal; momentum and the directional lateral response are odd.

For two opposite internal traversal states

$$q_{\text{dot_plus}} = q_{\text{dot_0}} + q_{\text{dot_v}} \quad q_{\text{dot_minus}} = -q_{\text{dot_0}} + q_{\text{dot_v}}$$

the quadratic scalar obeys

$$\Phi(q_{\text{dot_plus}}) + \Phi(q_{\text{dot_minus}}) = 2 \Phi(q_{\text{dot_0}}) + 2 \Phi(q_{\text{dot_v}})$$

The cross terms cancel exactly. The summed phase energy is a constant internal counter-flow energy plus a quadratic translational energy. This is the mathematical bridge between balanced counterflow and ordinary inertial curvature.

Exact cancellation of counter-flow cross terms

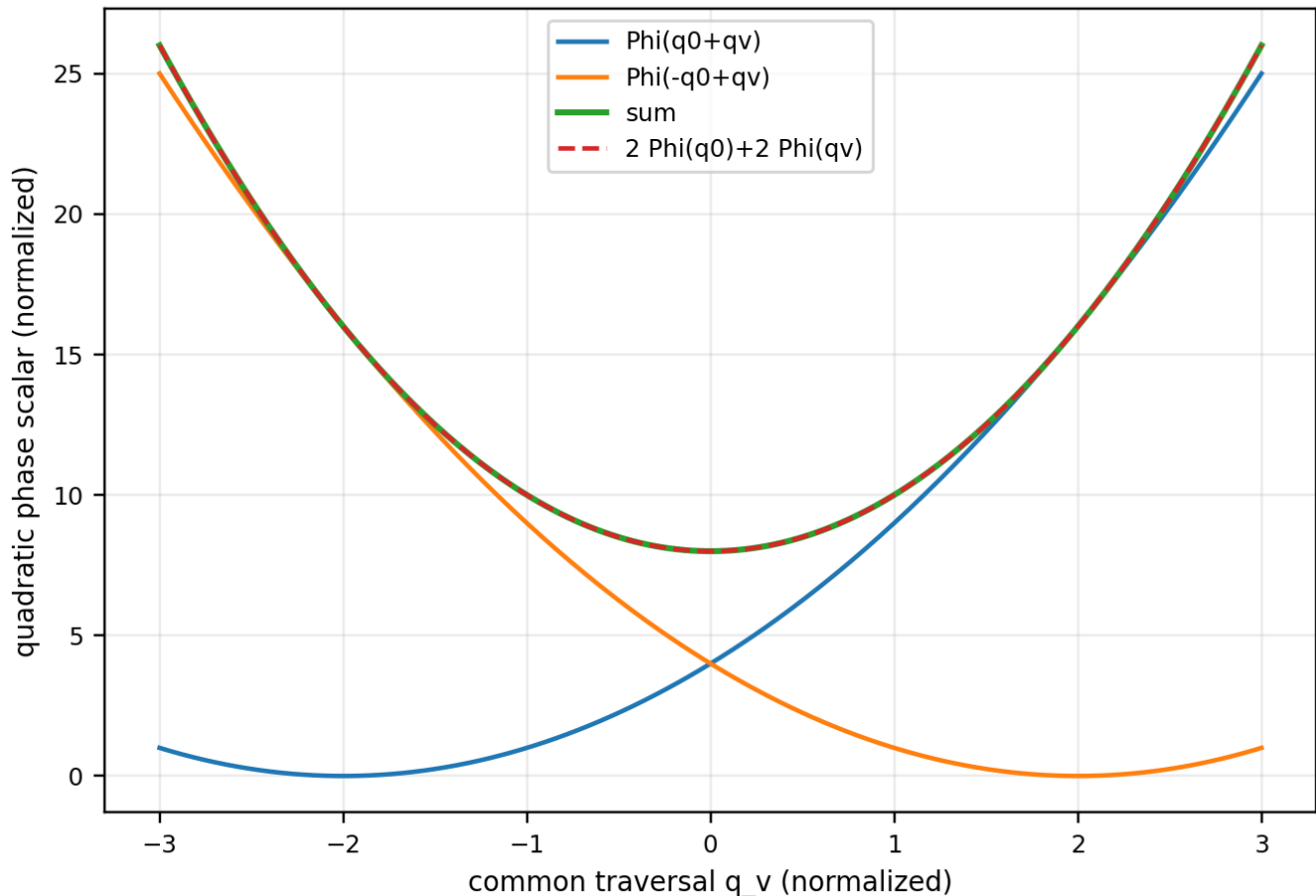


Figure 11. Direct numerical demonstration of exact cross-term cancellation for a normalized quadratic scalar with $q_0 = 2$.

9.2 Differential-pair equations

$$F_K/C_K = m_e \omega_e - m_p |\omega_p| \quad L = m_e \rho_e^2 \omega_e - m_p \rho_p^2 |\omega_p|$$

The first observable is weighted by $m \omega$. The second is weighted by $m \rho^2 \omega$. They are linearly independent whenever ρ_e^2 and ρ_p^2 are unequal after any shape factors are included. Therefore zero total angular momentum does not require zero lateral response, and zero lateral response does not require zero total angular momentum.

Let $R = \rho_p/\rho_e$. Three balance states follow:

Balance state	Condition	Physical interpretation
K-response null	$\omega_e/ \omega_p = \eta$	equal $m \omega$ weighting; equal phase-gradient state when phase radii are normalized
equal rotational energy	$\omega_e/ \omega_p = \sqrt{\eta} R$	equal even energy, generally nonzero angular momentum

Balance state	Condition	Physical interpretation
angular-momentum null	$\omega_e / \omega_p = \eta R^2$	equal and opposite $m \rho^2 \omega$ weighting; generally nonzero K response

If reciprocal Compton wavenumbers define the phase radii, $R = \eta$. The three frequency ratios then become η , $\eta^{3/2}$, and η^3 . Their large separation proves that a single literal mechanical shell picture cannot simultaneously represent energy balance, angular-momentum balance, and K-response balance. The phase-space interpretation is required.

At a common normalized phase radius, the K-response null $m_e \omega_e = m_p |\omega_p|$ is also an equal momentum condition. Equal momenta imply equal de Broglie wavelengths. Thus the $m \omega$ cancellation channel can be read as an equal-wavelength or equal-phase-gradient state.

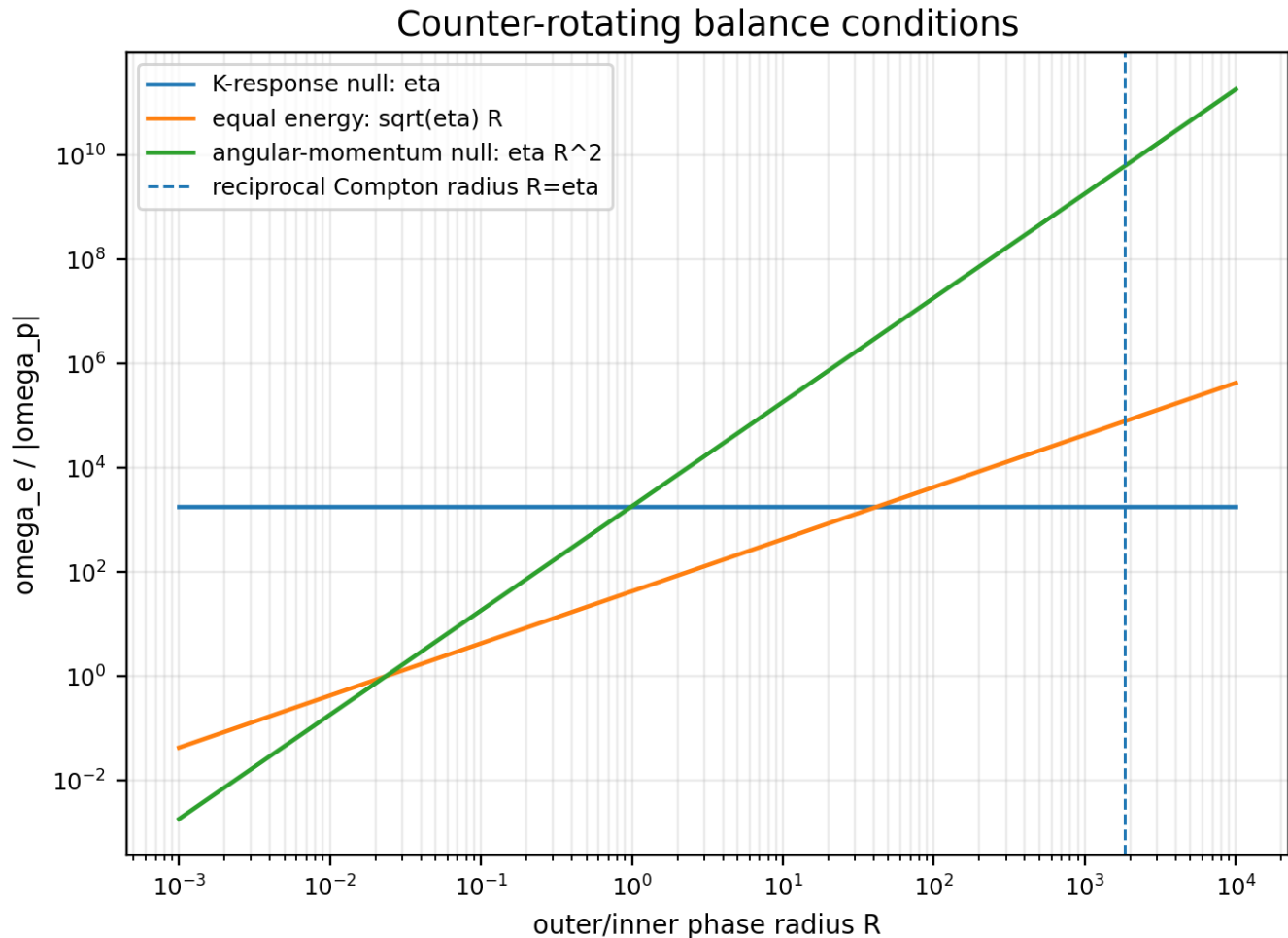


Figure 12. The three counter-rotating balance laws as functions of phase-radius ratio R . The vertical line marks $R = \eta$, the reciprocal-Compton ordering.

10. Earth-CMB vector and directional response

The corpus transformed Earth-CMB vector is

$$\mathbf{g}_K = (-880221.981635, -143911.436562, 266335.026949) \text{ m/s}$$

Its magnitude and unit direction are

$$|\mathbf{g}_K| = 930825.217270519235 \text{ m/s} \quad \hat{\mathbf{g}} = (-0.945636157362, -0.154606293310, +0.286127859460)$$

The lateral response direction is

F_K vector proportional to $\Omega \times g_K$

It reverses under spin reversal, vanishes when the rotor axis is parallel to g_K , and reaches maximum magnitude when the axis is perpendicular to g_K . The direction supplies an orientation-dependent projection, but a direction alone cannot generate a scalar mass ratio. A scalar closure additionally requires the speed β_E , the transformed metric magnitude or directional factor, a discrete mode index, and a normalization law.

Directional K-response geometry (unit vectors)

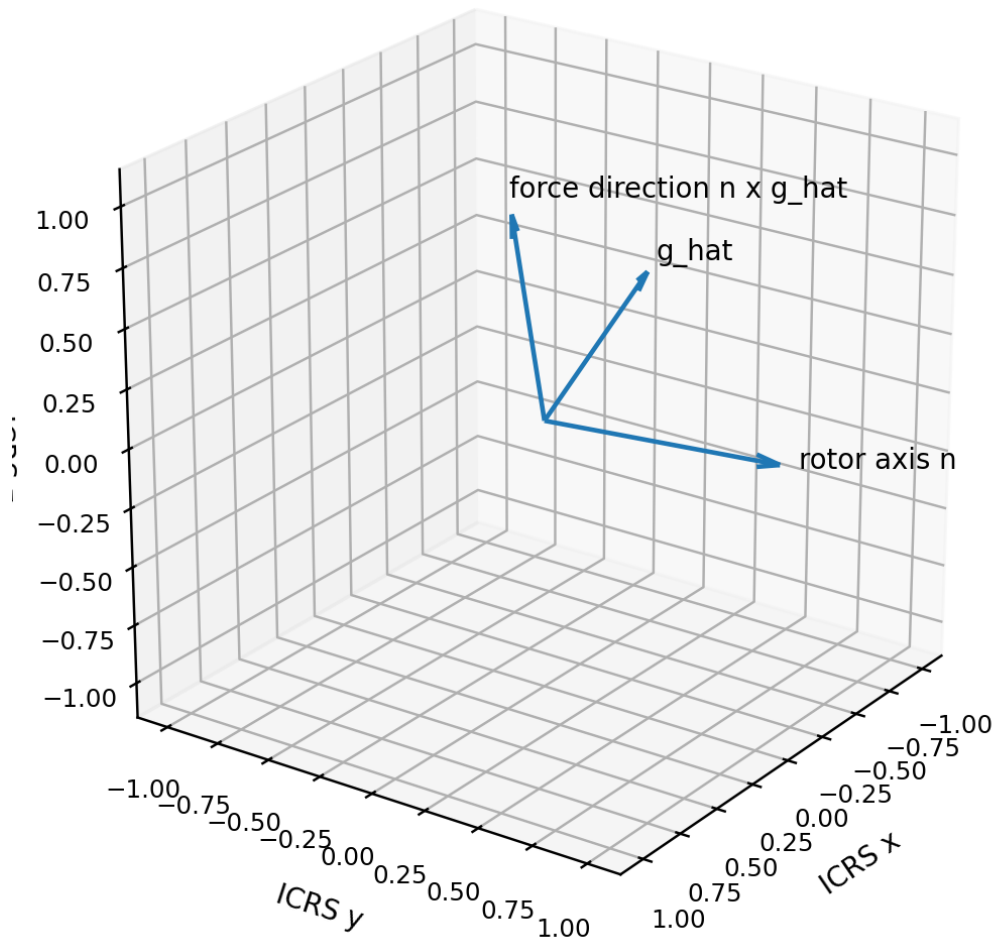


Figure 13. Unit-vector geometry of the transformed Earth-CMB direction, an orthogonal rotor axis, and the resulting n cross g force direction.

10.1 Secondary 10/9 observation

$$(|g_K|/V_E)/(\eta \beta_E) = 1.11173573377$$

This is within 56.216 parts per million of 10/9. Equivalently, η is near $(9/10)(|g_K|/V_E)/\beta_E$. The appearance of 10/9 may be related to the decade separation between the 1000 and 100 Mass-Bridge registries, but the present geometry does not independently select this factor. It is therefore a secondary clue rather than a primary bridge identity.

11. Exploratory two-body correction stack

The reported $\sqrt{11}$ CMB closure is high by approximately 296.041 parts per million. Counter-rotating equal-energy balance naturally introduces square-root mass factors. The two-body reduced mass is

$$\mu = m_e m_p / (m_e + m_p)$$

and the associated electron-normalized square-root factor is

$$\text{sqrt}(\mu/m_e) = \text{sqrt}(\eta/(\eta+1)) = 0.999727802667$$

Circular electron dynamics at $v = \alpha c$ supplies

$$\text{sqrt}(1-\alpha^2) = 0.999973373968$$

Applying both factors to the reported CMB closure gives

$$\eta_{\text{trial}} = \text{sqrt}(11) S_{\text{CMB}} \text{sqrt}(\mu/m_e) \text{sqrt}(1-\alpha^2) = 1836.147415828028$$

The residual is -2.863 parts per million. This is a much smaller residual than the raw CMB closure. It is not an independent derivation because $\text{sqrt}(\mu/m_e)$ contains η . Replacing η inside that factor with the raw geometric prediction gives 1836.147563723871, with nearly the same residual. The useful result is the structure of the correction stack, not a claim of algebraic closure.

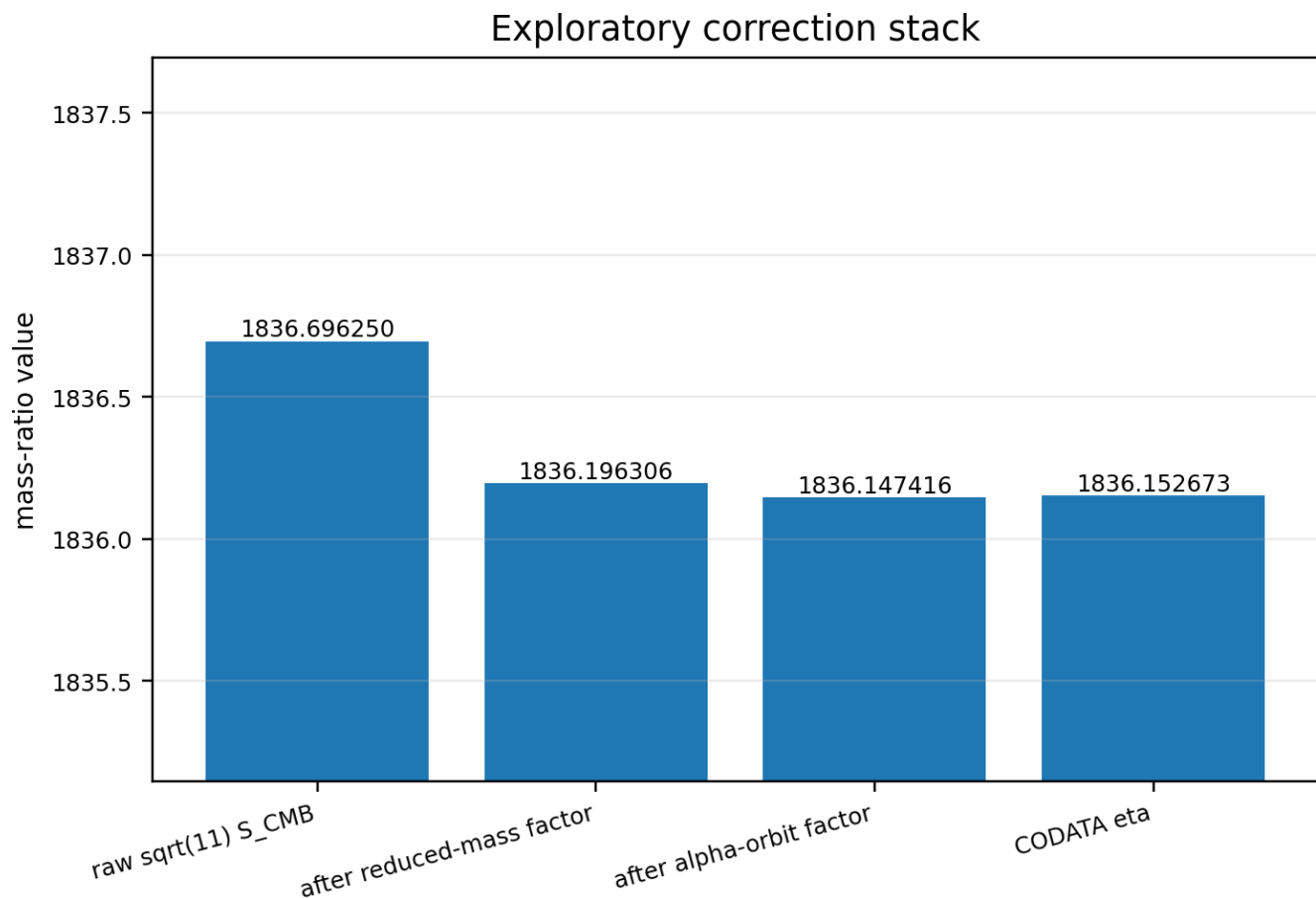


Figure 14. Numerical correction stack from the reported $\text{sqrt}(11)$ CMB value through the reduced-mass and alpha-orbit factors to the CODATA mass ratio.

12. Hidden geometric connections and derived consequences

12.1 One bridge, not two coincidences

Because $\eta\alpha$ is fixed by η and α , the cross-ratio creates a closure test. The close match of $10 A_{ac}/A_{ab}$ to $\eta\alpha$ means the two primary area ratios share a consistent internal geometry. This is stronger than matching η and α^{-1} separately because the common A_{bc} scale cancels.

12.2 The electromagnetic node is the bridge pivot

The classical electron radius is not an unrelated third length. It is the electron reduced Compton scale multiplied by alpha. It therefore converts the electron inertial phase scale into an electromagnetic scale. The proton reduced Compton scale then enters through eta. The triangle can be read as a sequence:

electron inertial phase scale --(alpha)--> electromagnetic radius --(eta alpha)--> proton inertial phase scale

12.3 Direct and reciprocal duality

Length ratios invert when converted to wavenumber ratios. The direct-space ordering $\lambda_{\bar{p}} < r_e < \lambda_{\bar{e}}$ becomes the reciprocal ordering $k_{C,e} < 1/r_e < k_{C,p}$. The requested inner-electron/outer-proton rotor order is exactly the reciprocal form of the known direct-space scale order.

12.4 CMB translation is common-mode before geometry

At one Earth-CMB velocity, λ_{dB}/λ_C is independent of mass. The electron and proton therefore acquire the same dimensionless translational phase ratio. The mass hierarchy remains entirely in the relative Compton scales. This supplies a clean decomposition into a universal external traversal and an internal Mass-Bridge geometry.

12.5 Counterrotation separates parity sectors

The quadratic phase energy survives reversal, while $m\omega$ and directed response reverse. A balanced pair can retain even internal energy while cancelling one odd channel and exposing another. This is why an inner/outer phase pair is a useful microscopic analogue even though it must not be interpreted as two conventional rigid rotors.

12.6 Experimental null architecture

The mathematical separation suggests a sequence of null states: angular-momentum null with nonzero lateral response; K-response null with nonzero angular momentum; double-null geometry; spin reversal; axis-orientation sweep; and radius-ratio sweep at fixed transferred angular momentum. Each operation changes one model variable while preserving major common-mode disturbances.

13. Consolidated conclusions

The Cell3 face areas define a precise three-ratio structure. The primary ratios reproduce eta and α^{-1} at approximately 8.6 parts per billion, while their shared-face-independent cross-ratio reproduces eta alpha at approximately 89 parts per trillion. A single common A_{bc} scale correction brings the two primary ratios into equal-and-opposite agreement at roughly 4.45 times 10^{-11} relative error.

The corresponding known-physics scales are the electron reduced Compton wavelength, classical electron radius, and proton reduced Compton wavelength. Their pairwise ratios obey exact identities. The Mass-Bridge is therefore projectively equivalent to one closed Compton-classical-radius triangle.

The reduced Compton wavelength identifies the closed angular phase sector; the ordinary Compton and de Broglie wavelengths identify linear phase closure. The Earth-CMB speed supplies a mass-independent de Broglie-to-Compton factor shared by electron and proton. The K-Field directional factor and $\sqrt{11}$ mode move this common factor into the proton-electron mass-ratio neighborhood.

Counterrotation supplies the parity mechanism. Even quadratic energy, odd angular momentum, and odd lateral response are different functionals of the same internal motion. Their distinct $m\omega$ and $m\rho^2\omega$ weightings explain why energy balance, angular-momentum balance, and lateral-response balance occur at different frequency ratios.

Archival synthesis

The central architecture is:

electron reduced Compton scale - classical electron radius - proton reduced Compton scale;

**universal Earth-CMB traversal - K-Field directional projection - discrete mode registry;
balanced counterflow energy - differential momentum - differential lateral response.**

Appendix A. Complete numerical constant table

Name	Symbol or definition	Value	Unit
speed of light	c	2.997924580000000e+08	m/s
Planck constant	h	6.626070150000000e-34	J s
reduced Planck constant	$\hbar$	1.054571817646156e-34	J s
electron mass	m_e	9.109383713900000e-31	kg
proton mass	m_p	1.672621925950000e-27	kg
fine-structure constant	α	7.297352564300000e-03	1
inverse fine-structure constant	α^{-1}	1.370359991775901e+02	1
proton-electron mass ratio	η	1.836152673426000e+03	1
classical electron radius	r_e	2.817940320500000e-15	m
electron reduced Compton	$\hbar/(m_e c)$	3.861592674352375e-13	m
proton reduced Compton	$\hbar/(m_p c)$	2.103089100513957e-16	m
electron ordinary Compton	$h/(m_e c)$	2.426310235380317e-12	m
proton ordinary Compton	$h/(m_p c)$	1.321409853603883e-15	m
Earth-CMB speed	V_E	3.698273920000000e+05	m/s
Earth-CMB beta	V_E/c	1.233611393919723e-03	1
Earth-CMB gamma	$1/\sqrt{1-\beta^2}$	1.000000760899404e+00	1
electron CMB de Broglie	$h/(\gamma m_e V_E)$	1.966833640774237e-09	m
proton CMB de Broglie	$h/(\gamma m_p V_E)$	1.071171079205084e-12	m

Appendix B. Complete 27-node coordinate table

Coordinates are computed as $(i a + j b + k c)/2$ in the canonical reconstruction. The physical columns use 16.0686029005 nm per native unit.

Address	Class	x native	y native	z native	x nm	y nm	z nm
-1,-1,-1	corner	-0.016418080	-0.033526108	-0.018828574	-0.263815602	-0.538717712	-0.302548872
-1,-1,+0	edge center	-0.012326937	-0.018144917	+0.000000000	-0.198076662	-0.291563458	+0.000000000
-1,-1,+1	corner	-0.008235795	-0.002763725	+0.018828574	-0.132337723	-0.044409205	+0.302548872
-1,+0,-1	edge center	-0.014489672	-0.015381191	-0.018828574	-0.232828778	-0.247154253	-0.302548872
-1,+0,+0	face center	-0.010398529	+0.000000000	+0.000000000	-0.167089839	+0.000000000	+0.000000000
-1,+0,+1	edge center	-0.006307387	+0.015381191	+0.018828574	-0.101350900	+0.247154253	+0.302548872
-1,+1,-1	corner	-0.012561263	+0.002763725	-0.018828574	-0.201841955	+0.044409205	-0.302548872
-1,+1,+0	edge center	-0.008470121	+0.018144917	+0.000000000	-0.136103016	+0.291563458	+0.000000000
-1,+1,+1	corner	-0.004378979	+0.033526108	+0.018828574	-0.070364076	+0.538717712	+0.302548872
+0,-1,-1	edge center	-0.006019550	-0.033526108	-0.018828574	-0.096725763	-0.538717712	-0.302548872
+0,-1,+0	face center	-0.001928408	-0.018144917	+0.000000000	-0.030986823	-0.291563458	+0.000000000
+0,-1,+1	edge center	+0.002162734	-0.002763725	+0.018828574	+0.034752116	-0.044409205	+0.302548872
+0,+0,-1	face center	-0.004091142	-0.015381191	-0.018828574	-0.065738939	-0.247154253	-0.302548872
+0,+0,+0	body center	+0.000000000	+0.000000000	+0.000000000	+0.000000000	+0.000000000	+0.000000000
+0,+0,+1	face center	+0.004091142	+0.015381191	+0.018828574	+0.065738939	+0.247154253	+0.302548872
+0,+1,-1	edge center	-0.002162734	+0.002763725	-0.018828574	-0.034752116	+0.044409205	-0.302548872
+0,+1,+0	face center	+0.001928408	+0.018144917	+0.000000000	+0.030986823	+0.291563458	+0.000000000
+0,+1,+1	edge center	+0.006019550	+0.033526108	+0.018828574	+0.096725763	+0.538717712	+0.302548872
+1,-1,-1	corner	+0.004378979	-0.033526108	-0.018828574	+0.070364076	-0.538717712	-0.302548872
+1,-1,+0	edge center	+0.008470121	-0.018144917	+0.000000000	+0.136103016	-0.291563458	+0.000000000
+1,-1,+1	corner	+0.012561263	-0.002763725	+0.018828574	+0.201841955	-0.044409205	+0.302548872
+1,+0,-1	edge center	+0.006307387	-0.015381191	-0.018828574	+0.101350900	-0.247154253	-0.302548872
+1,+0,+0	face center	+0.010398529	+0.000000000	+0.000000000	+0.167089839	+0.000000000	+0.000000000
+1,+0,+1	edge center	+0.014489672	+0.015381191	+0.018828574	+0.232828778	+0.247154253	+0.302548872
+1,+1,-1	corner	+0.008235795	+0.002763725	-0.018828574	+0.132337723	+0.044409205	-0.302548872
+1,+1,+0	edge center	+0.012326937	+0.018144917	+0.000000000	+0.198076662	+0.291563458	+0.000000000
+1,+1,+1	corner	+0.016418080	+0.033526108	+0.018828574	+0.263815602	+0.538717712	+0.302548872

Appendix C. Compact derivation set

C.1 Known-physics triangle

$\lambda_{\bar{e}} = \hbar / (m_e c)$ $\lambda_{\bar{p}} = \hbar / (m_p c)$ $r_e = \alpha \lambda_{\bar{e}}$ therefore
 $\lambda_{\bar{e}} / \lambda_{\bar{p}} = \eta$ therefore $\lambda_{\bar{e}} / r_e = \alpha^{-1}$ therefore
 $r_e / \lambda_{\bar{p}} = \eta \alpha$

C.2 Mass-Bridge triangle

$\eta_{\text{geom}} = 1000 A_{bc} / A_{ab}$ $\alpha_{\text{inv_geom}} = 100 A_{bc} / A_{ac}$ $\chi_{\text{geom}} = 10 A_{ac} / A_{ab} =$
 $\eta_{\text{geom}} / \alpha_{\text{inv_geom}}$

C.3 Earth-CMB common factor

$\lambda_{\text{dB}} / \lambda_{\text{C}} = [h / (\gamma m v)] / [h / (m c)] = 1 / (\gamma \beta)$

C.4 Counter-rotating balance

$F_K / C_K = m_e \omega_e - m_p |\omega_p|$ $L = m_e \rho_e^2 \omega_e - m_p \rho_p^2 |\omega_p|$ $K \text{ null:}$
 $\omega_e / |\omega_p| = \eta$ equal energy: $\omega_e / |\omega_p| = \sqrt{\eta}$ $R \text{ null:}$
 $\omega_e / |\omega_p| = \eta R^2$

C.5 Quadratic parity cancellation

$(q_0 + q_v)^T W (q_0 + q_v) + (-q_0 + q_v)^T W (-q_0 + q_v) = 2 q_0^T W q_0 + 2 q_v^T W q_v$

Appendix D. Reproducibility record

The companion JSON file contains every input constant, canonical cell vector, plane normal, vertex, 27-node coordinate, bridge ratio, residual, Earth-CMB wavelength, counter-rotor condition, exploratory correction factor, and plotted parity sample. The JSON schema identifier is `kfield.compton_classical_radius_mass_bridge_triangle.v1`.

PDF filename: `K-Field_Compton_classical_radius_Mass_Bridge_triangle_1785175483.pdf`
 JSON filename: `K-Field_Compton_classical_radius_Mass_Bridge_triangle_1785175483.json`
 Epoch: 1785175483
 Generated UTC: 2026-07-27T18:04:43.208974Z
 SciPy version: 1.17.0
 Numerical payload SHA-256: `e5cbf8ae467225c6df01e27705e19ea0446cfa799719003524f1299dd75eee96`

All figures are generated from the machine-readable numerical payload. The PDF uses only the built-in ReportLab core text fonts Helvetica, Helvetica-Bold, Times-Roman, and Courier. No external TrueType font is embedded.

References

1. National Institute of Standards and Technology. 2022 CODATA Recommended Values of the Fundamental Physical Constants. Values accessed through the SciPy 1.17.0 `scipy.constants` table.
2. L. de Broglie. *Recherches sur la theorie des quanta*. Doctoral thesis, University of Paris, 1924.
3. A. H. Compton. A quantum theory of the scattering of X-rays by light elements. *Physical Review* 21, 483-502, 1923.
4. Krytronx, LLC K-Field Cell3 geometry, Earth-CMB vector, Mass-Bridge area data, linear-inertia phase-energy equations, and concentric counter-rotor differential equations, 2026 archival corpus.